

Cosmic-Ray Neutron Sensing (CRNS) for Dual Landslide & Flood Early Warning Systems

Executive Summary

Slope failures (landslides) and catchment inundations (floods) have historically been treated as isolated geohazards with disjoint monitoring systems. However, both hazards share the exact same physical driver: **antecedent root-zone and vadose-zone soil saturation**.

- **Point-scale sensors** (TDR, FDR, capacitance probes) suffer from extreme spatial representation error in heterogeneous soil and require invasive borehole drilling that destroys delicate slip surfaces.
- **Satellite remote sensing** (SMAP, Sentinel-1 SAR) cannot penetrate dense forest canopy and has coarse revisit cycles (days) incompatible with flash hazard dynamics.
- **Cosmic-Ray Neutron Sensing (CRNS)** bridges this scale gap: it measures area-averaged volumetric soil water content (θ) across an intermediate footprint of **130–240 m radius** and **15–70 cm depth**, completely non-invasively.

1. Physical Foundations of CRNS

1.1 Neutron Moderation by Hydrogen

Primary galactic cosmic rays (predominantly high-energy protons) collide with atmospheric nitrogen and oxygen nuclei, producing high-energy secondary neutrons. As fast neutrons enter the ground, they undergo elastic collisions with atomic nuclei.

Because hydrogen has an atomic mass virtually identical to a neutron, **neutrons lose ~50% of their kinetic energy in a single collision with a hydrogen nucleus** (water), compared to <2% with silicon or oxygen. Moderated neutrons bounce back (albedo) into the lower atmosphere:

$$\text{Above-ground epithermal neutron flux } N \propto \frac{1}{\theta_{\text{soil}}}$$

1.2 Footprint Geometry (URANOS Monte Carlo Modeling)

- **Horizontal Footprint Radius ($R_{\{86\}}$):** Spans **130 m to 240 m** around the probe (covering 5 to 18 hectares).
- **Vertical Sensing Depth ($D_{\{86\}}$):** Spans **15 cm** in saturated soils to **70 cm** in arid conditions:

$$D_{\{86\}} = \frac{1}{\rho_{\text{bulk}}} \leq ft(a_0 + a_1 \cdot \theta \right)^{-1}$$

- **Non-Invasive Hillslope Integrity:** The probe sits on a surface mast, preserving intact slope stratigraphy and natural drainage paths.

1.3 Calibration & Environmental Corrections

Raw neutron counts (N_{raw}) are corrected for atmospheric barometric pressure (f_p), ambient water vapor (f_v), solar cosmic flux variations (f_i), and vegetation biomass (f_{bio}):

$$N_{\text{corr}} = N_{\text{raw}} \cdot f_p \cdot f_v \cdot f_i \cdot f_{\text{bio}}$$

Soil moisture is then computed via the **Desilets Universal Formulation**:

$$\theta = \left[\frac{a_0}{\frac{N_{\text{corr}}}{N_0} - a_1} - a_2 \right] \cdot \frac{\rho_{\text{bulk}}}{\rho_w} - \frac{w_{\text{lattice}} - w_{\text{SOC}}}{\rho_w}$$

Where $a_0 = 0.0808$, $a_1 = 0.372$, $a_2 = 0.115$, and N_0 is the site-specific calibration constant (count rate over dry silica).

2. Landslide Detection & Geotechnical Modeling

2.1 Failure Mechanism

Rainfall induces shallow translational landslides when water infiltration increases pore-water pressure (u), reducing effective stress ($\sigma' = \sigma - u$). According to the Mohr-Coulomb failure criterion:

$$\tau_f = c' + (\sigma - u) \tan \phi'$$

As CRNS soil moisture approaches saturation ($\theta \rightarrow \theta_{\text{sat}}$), suction stress vanishes, pore-water pressure spikes, and the **Factor of Safety (FS)** collapses below 1.0:

$$FS = \frac{c' + (\gamma_{\text{bulk}} \cdot z \cdot \cos^2 \beta - u) \tan \phi'}{\gamma_{\text{bulk}} \cdot z \cdot \sin \beta \cos \beta}$$

2.2 Dynamic Rainfall Intensity-Duration (I-D) Threshold Shifting

Standard empirical thresholds ($I = \alpha \cdot D^{-\beta}$) fail because a 50 mm storm falling on dry soil produces zero movement, whereas the same storm on saturated soil causes catastrophic failure. We update the threshold dynamically using CRNS antecedent moisture:

$$\alpha(\theta) = \alpha_{\text{dry}} \cdot \left[1 - \frac{\theta}{\theta_{\text{sat}}} \right]^{\gamma}$$

3. Flood Prediction Extension ("Based on Something Similar")

3.1 Hydrological Coupling: Horton vs. Dunne Runoff

- **Hortonian Runoff (Infiltration Excess):** Occurs when rainfall rate exceeds infiltration capacity.
- **Dunne Runoff (Saturation Excess):** Occurs in humid and mountain catchments when the soil column is 100% saturated. Rain cannot infiltrate and immediately transforms into surface runoff.

3.2 Dynamic SCS-CN Catchment Runoff Formulation

The potential maximum soil water retention S_{ret} is updated in real time by the CRNS moisture deficit:

$$S_{ret}(t) = S_{max} \cdot \le ft[1 - \frac{\theta_{CRNS}(t)}{\theta_{sat}} \right]$$

Direct runoff depth (Q_{excess}) is calculated as:

$$Q_{excess} = \frac{(P - I_a)^2}{P - I_a + S_{ret}(t)} \text{ for } P > I_a = 0.15 \cdot S_{ret}(t)$$

Streamflow discharge $Q(t)$ at the downstream catchment outlet is forecasted via synthetic unit hydrograph convolution:

$$Q(t) = Q_{base} + \sum_{\tau=0}^t Q_{excess}(\tau) \cdot u(t - \tau)$$

This provides **6 to 48 hours of actionable evacuation lead time** for civil protection authorities.

4. Operational Early Warning Matrix

Alert Level	Landslide FS	CRNS Saturation (θ / θ_{sat})	Downstream Q (m^3/s)	Recommended Actions
GREEN (Normal)	$FS > 1.50$	< 0.65	< 15.0	Routine 1-hour telemetry polling; standard logging.
YELLOW (Advisory)	$1.25 < FS \leq 1.50$	$0.65 \leq S_r < 0.80$	$15.0 - 30.0$	Increase polling to 15 mins; inspect culverts & gullies.
ORANGE (Warning)	$1.00 < FS \leq 1.25$	$0.80 \leq S_r < 0.92$	$30.0 - 45.0$	Close vulnerable mountain roads; mobilize first responders.
RED (Imminent)	$FS \leq 1.00$	≥ 0.92	> 45.0	Sound sirens, send automated cell broadcasts, mandatory evacuation.

5. Artifacts and Generated Files

The generated PDF and executable Python pipeline have been compiled:

- **Comprehensive PDF Report:** [CRNS Landslide and Flood Prediction Report.pdf](#)
- **Standalone Python Pipeline:** [crns_hazard_pipeline.py](#)
- **High-Resolution Figures:**
 - [crns_mechanism.png](#)
 - [simulation_results.png](#)

Embedded Scientific Figures & Simulation Results

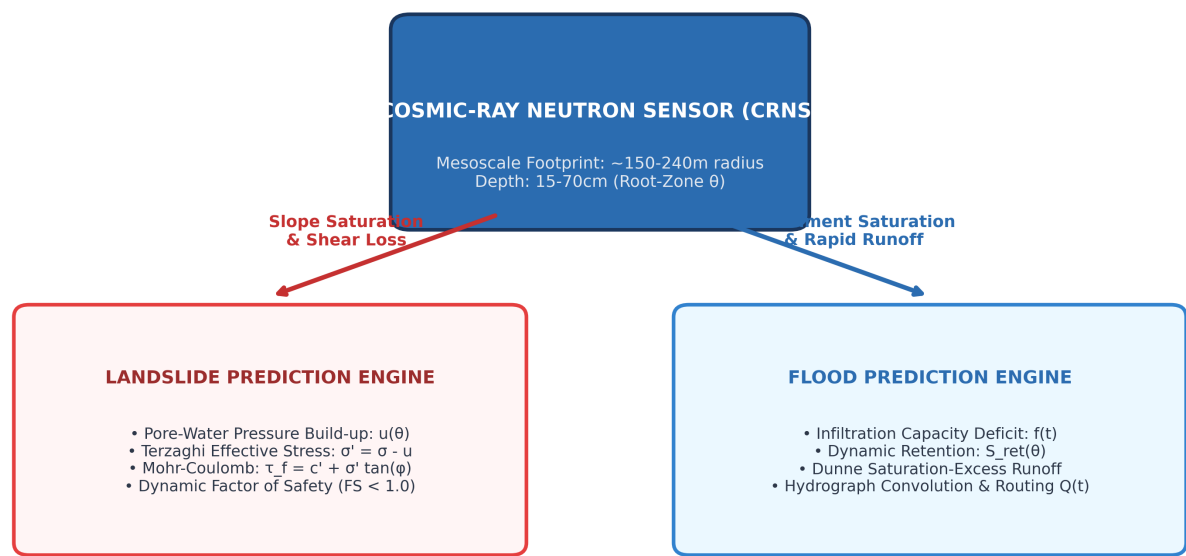


Figure 1: Cosmic-Ray Neutron Sensing physics and dual geohazard coupling mechanism.

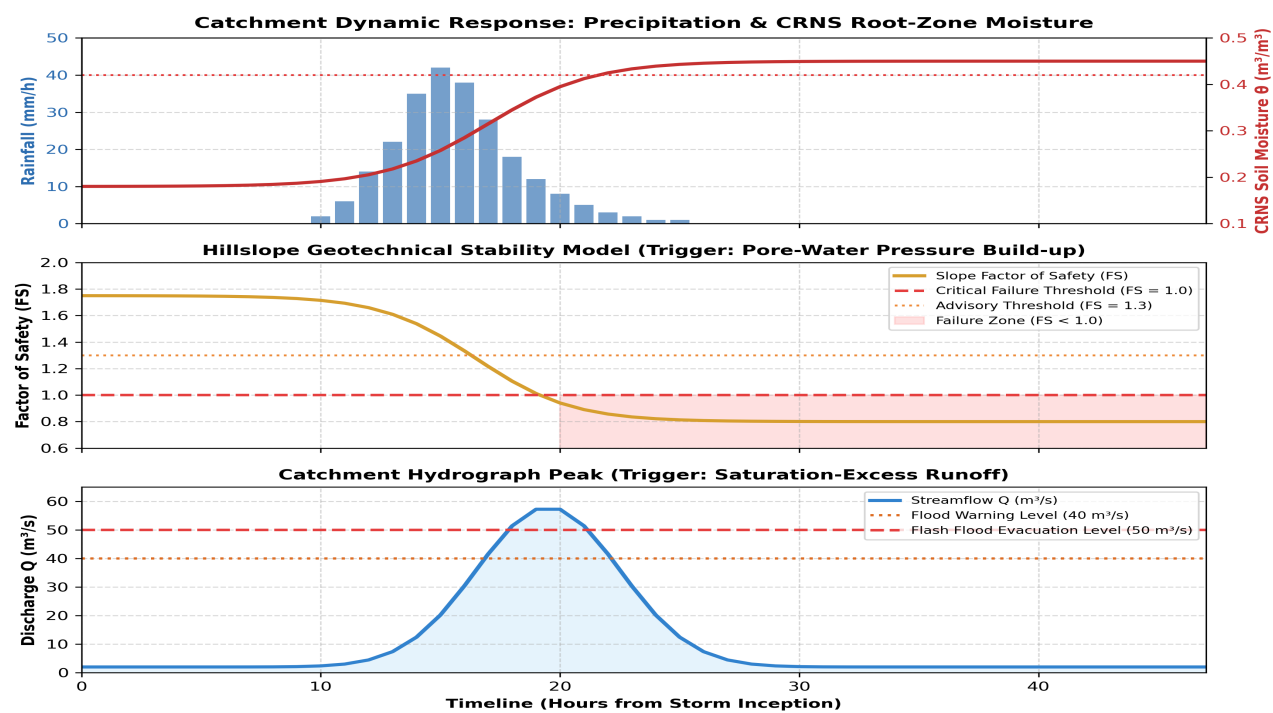


Figure 2: 48-Hour storm response simulation: Soil moisture, Landslide FS, and Flood Hydrograph.